

June 22, 2021



U.S. Department
of Transportation

East Building, PHH-30
1200 New Jersey Avenue S.E.
Washington, D.C. 20590

**Pipeline and Hazardous
Materials Safety Administration**

DOT-SP 21239

EXPIRATION DATE: 2021-12-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Lockheed Martin Corporation
Littleton, Co
2. PURPOSE AND LIMITATIONS:
 - a. This emergency special permit authorizes the transportation in commerce of low production lithium ion batteries packed with equipment in alternative packaging by ground transportation. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.185(e)(3) in that the outer packaging is not a UN-Specification packaging, except as specified herein.
5. BASIS: This special permit is based on the application of Lockheed Martin Corporation dated June 9, 2021 and

June 22, 2021

supplemental information dated June 14, 2021, submitted in accordance with § 107.105 and the public proceeding thereon.

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Material Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Lithium ion batteries packed with equipment <i>including lithium ion polymer batteries</i>	9	UN3481	N/A

7. SAFETY CONTROL MEASURES:

a. OPERATIONAL CONTROLS:

(1) This special permit is valid only for transportation of lithium ion batteries packed with equipment (The installed battery is electrically disconnected from the spacecraft during shipment) by ground.

(2) The battery is affixed to a spacecraft bus and consists of two battery blocks with each block made of 64 component cells (identified as ICR-18650M) connected in a configuration as described in the Lockheed Martin's June 9, 2021 application and on file with the Office of Hazardous Materials Safety (OHMS).

(3) The battery has a maximum rated energy capacity of 1480 Wh.

(4) The battery must be equipped with a means of preventing dangerous reverse current flow for cells and blocks connected in parallel.

June 22, 2021

(5) Not more than one spacecraft is permitted within a package.

(6) The spacecraft must be constructed to prevent against accidental operation during transport.

(7) Battery terminals must be protected from short-circuiting.

b. TESTING REQUIREMENTS:

(1) Each cell in the battery must be a type that has passed the required tests in accordance with the UN Manual of Tests and Criteria, 6th Revised Edition.

(2) The battery packed with equipment must have been designed to operate in the extreme environment of outer space (i.e., burst test under extreme conditions).

c. PACKAGING:

(1) The battery installed in but electrically disconnected from the spacecraft during shipment must be securely affixed to the spacecraft to prevent the movement of the battery.

(2) The spacecraft and the spacecraft electrical bus must have been designed to handle the rigors of operating in outer space.

(3) The spacecraft must be placed and secured in a non-specification enclosed metal, wood or fiberglass van trailers or freight shipping container (identified as EPL6305119).

(4) The shipping container must be provided a means of minimizing shock and vibration during road transportation.

(5) The container bases must be secured to vibration dampers to withstand road inputs.

(6) The net weight of the lithium ion battery on the spacecraft may not exceed 8 kg.

d. MARKING REQUIREMENTS: Each package prepared under the provisions of this special permit must be plainly marked with the permit number "DOT-SP 21239".

8. SPECIAL PROVISIONS:

a. A person who is not a holder of this special permit, but receives a packaging covered by this special permit, may reoffer it for transportation provided no modification or change is made to the packaging and it is offered for transportation in conformance with this special permit and the HMR.

b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each motor vehicle used to transport packages covered by this special permit.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive

June 22, 2021

training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)–“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-30, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at

June 22, 2021

<https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Steve H/Andrew Eckenrode